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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 27	
TITLE: Develop a Java application demonstrating socket programming for communication between a client and a server.	

Problem Statement

Develop a Java application demonstrating socket programming for communication between a client and a server.

Learning Objectives

To establish communication between client and server using Java sockets.

Programme

1.Create a simple client-server chat application using sockets.

Server Code:

```
import java.io.*;
import java.net.*;

public class SocketServer1 {
   public static void main(String[] args) {
      try {
         ServerSocket server = new ServerSocket(5000);
         System.out.println("Server waiting for client...");

         Socket socket = server.accept();
         System.out.println("Client connected.");

         BufferedReader input = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

         PrintWriter output = new PrintWriter(
            socket.getOutputStream(), true);

         BufferedReader keyboard = new BufferedReader(
            new InputStreamReader(System.in));
```

```
            String message;

            while (true) {
               message = input.readLine();

               if (message == null || message.equalsIgnoreCase("exit")) {
                  break;
               }

               System.out.println("Client: " + message);

               System.out.print("Server reply: ");
               message = keyboard.readLine();

               output.println(message);

               if (message.equalsIgnoreCase("exit")) {
                  break;
               }
            }

            socket.close();
            server.close();
         } catch (IOException e) {
            System.out.println(e.getMessage());
         }
      }
   }
```

Client Code:

```
import java.io.*;
import java.net.*;

public class SocketClient1 {
   public static void main(String[] args) {
      try {
         Socket socket = new Socket("localhost", 5000);
         System.out.println("Connected to server.");

         BufferedReader input = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

         PrintWriter output = new PrintWriter(
```

```
            socket.getOutputStream(), true);

        BufferedReader keyboard = new BufferedReader(
            new InputStreamReader(System.in));

        String message;

        while (true) {
            System.out.print("Enter message: ");
            message = keyboard.readLine();

            output.println(message);

            if (message.equalsIgnoreCase("exit")) {
                break;
            }

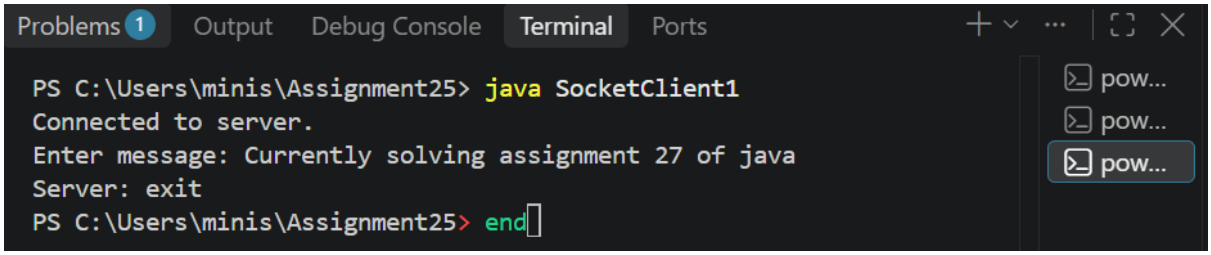
            message = input.readLine();
            System.out.println("Server: " + message);

            if (message == null || message.equalsIgnoreCase("exit")) {
                break;
            }
        }

        socket.close();
    } catch (IOException e) {
        System.out.println(e.getMessage());
    }
  }
}
```

Output:





2. Create a Java program using socket programming where the client sends a message to the server, and the server returns the same message back (echo system).

Server Code:

```
import java.io.*;
import java.net.*;

public class EchoServer {
   public static void main(String[] args) {
      try {
         ServerSocket server = new ServerSocket(5001);
         System.out.println("Echo server waiting for client...");

         Socket socket = server.accept();
         System.out.println("Client connected.");

         BufferedReader input = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

         PrintWriter output = new PrintWriter(
            socket.getOutputStream(), true);

         String message = input.readLine();

         System.out.println("Received: " + message);
         output.println(message);

         socket.close();
         server.close();
      } catch (IOException e) {
         System.out.println(e.getMessage());
      }
   }
}
```

Client Code:

```
import java.io.*;
import java.net.*;

public class EchoClient {
   public static void main(String[] args) {
      try {
         Socket socket = new Socket("localhost", 5001);

         BufferedReader input = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));

         PrintWriter output = new PrintWriter(
            socket.getOutputStream(), true);

         BufferedReader keyboard = new BufferedReader(
            new InputStreamReader(System.in));

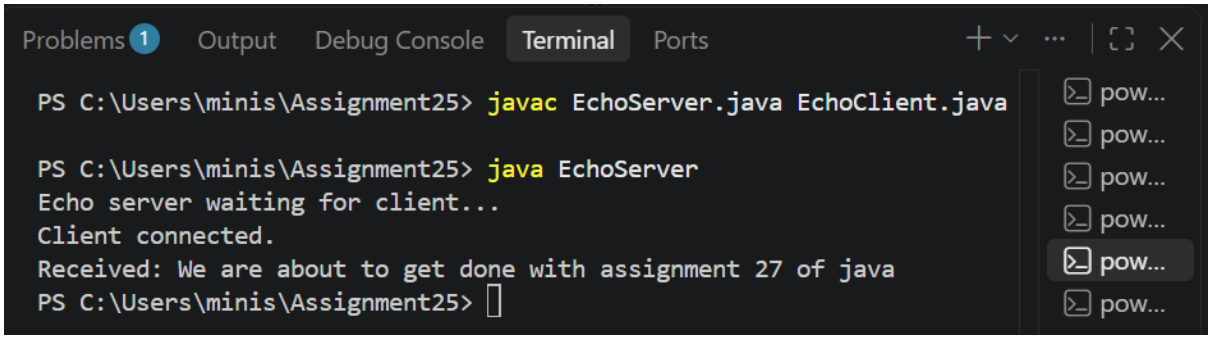
         System.out.print("Enter a message: ");
         String message = keyboard.readLine();

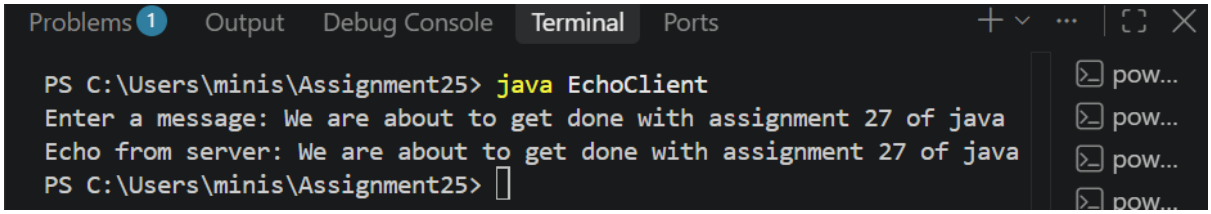
         output.println(message);

         String reply = input.readLine();
         System.out.println("Echo from server: " + reply);

         socket.close();
      } catch (IOException e) {
         System.out.println(e.getMessage());
      }
   }
}
```

Output:





Conclusion

This assignment helped us understand socket programming in Java and how a client and server communicate over a network. We learned to use ServerSocket to accept client connections and Socket to establish communication. We implemented a simple chat application to exchange messages and an echo system in which the server returns the client's message. Overall, this assignment improved our understanding of network communication and client-server applications in Java.